Week 6

Antiderivatives

the initial value problem (IVP)

$$\begin{cases} y' = x - 2 \\ y(0) = 3 \end{cases} \qquad y(x) = \int (x-2)\,dx$$

$$y(x) = \frac{x^2}{2} - 2x + C \quad \Rightarrow y(0) = C = 3$$

$$\Rightarrow y(x) = \frac{x^2}{2} - 2x + 3 \;//$$

Ex $\begin{cases} y' = \sec^2 x \\ y(0) = 1 \end{cases} \Rightarrow y(x) = \tan x + C$

$y(0) = C = 1$

$$\Rightarrow y(x) = \tan x + 1$$

Ex $\begin{cases} y' = \frac{1}{x^2} - \frac{1}{x^3} \\ y(-1) = 0 \end{cases}$

$$y(x) = \int \left(\frac{1}{x^2} - \frac{1}{x^3}\right) dx$$

$$= \int \left(x^{-2} - x^{-3}\right) dx$$

$$= -x^{-1} + \frac{x^{-2}}{2} + C$$

$$y(x) = -\frac{1}{x} + \frac{1}{2x^2} + C$$

Check by differentiate Answer

$$y(-1) = 1 + \frac{1}{2} + C = 0 \Rightarrow C = -\frac{3}{2}$$

$$\Rightarrow y(x) = -\frac{1}{x} + \frac{1}{2x^2} - \frac{3}{2}$$

Ex $\begin{cases} y'' = x^{-4} \\ y' = 2 \\ y(1) = 1 \end{cases}$

$$y'(x) = \int x^{-4}\,dx$$

$$= -\frac{x^{-3}}{3} + C = -\frac{1}{3x^3} + C$$

$$y'(1) = -\frac{1}{3} + C = 2$$

$$\Rightarrow C = \frac{7}{3} \quad \Rightarrow y'(x) = -\frac{1}{3x^3} + \frac{7}{3}$$

$$y(x) = \int \left(-\frac{x^{-3}}{3} + \frac{7}{3}\right) dx = \frac{x^{-2}}{6} + \frac{7}{3}x + D$$

$$y(1) = \frac{1}{6} + \frac{7}{3} + D = 1 \Rightarrow D = 1 - \frac{7}{3} - \frac{1}{6}$$

$$\Rightarrow D = \frac{6 - 14 - 1}{6} = -\frac{9}{6} = -\frac{3}{2}$$

$$y(x) = \frac{1}{6x^2} + \frac{7}{3}x - \frac{3}{2}$$

Midterm ↑

inverse function

Def f is called one-to-one

if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$